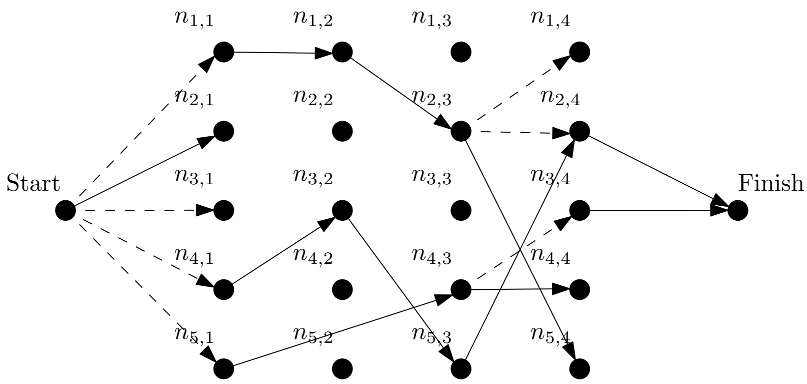
COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 7

1 Artificial Intelligence (sbh11)

Evil Robot has decided to toy with his victims by placing them in a puzzle maze. Victims are initially at the `Start` node and need to navigate to the `Finish` node. Other nodes $n_{i,j}$ are laid out on a grid and connected by two kinds of edge. Fixed one-way edges (single-source, solid lines) allow a victim to pass from one node to another, left to right. For example, $n_{5,1}$ to $n_{4,3}$ in the diagram. Switched edges (single-source, solid and dotted lines) can be moved among destinations. For example, $n_{2,3}$ is currently connected to $n_{5,4}$ but can be moved to connect to $n_{1,4}$ or $n_{2,4}$ instead.



A victim can set the switched edges as desired, but only if they have reached the switch's source node. Their aim is to cross the maze successfully.

State-variable planning representation

(*a*) The victim decides to solve this problem by representing it as a planning problem using the *state-variable* representation. Give an example of how each of the following elements might appear in the representation.

(*i*) A *domain*. [1 mark]

Answer: An example would be the set $D = \{n_{i,j}\} \cup \{\texttt{Start}, \texttt{Finish}\}$.

(*ii*) A *rigid relation*. [2 marks]

Answer: An example would be $\texttt{fixed}(n, n')$ where $n, n' \in D$. One for each fixed edge.

(*iii*) A *function*, including a *state variable*, representing the destination of a switched edge. [3 marks]

Answer: An example would be $\texttt{switch}(n, s) : D \times S \to D$ where s is the state variable and the function maps node n on the left of a switch to the switch's destination in state s.

(*iv*) A *goal*. [1 mark]

Answer: An example would be $\texttt{at}(s) = \texttt{Finish}$, where $\texttt{at}$ denotes the victim's location in state s.

State-variable planning representation

(*b*) Explain in detail how actions allowing (1) the victim to move around the grid, and (2) the destination of a switched edge to be altered, might be implemented in the state-variable representation. [6 marks]

Answer: An example for (1) would be an action $\texttt{move}(n')$ with preconditions $\texttt{at}(s) = n$ and $\texttt{fixed}(\texttt{n}, \texttt{n}')$, and effect $\texttt{at}(s) = n'$. An example of (2) would be $\texttt{move\_switch}(n, n')$ denoting the action of moving the switch at n to have destination n'. To properly constrain this, we would need preconditions corresponding to rigid relations $\texttt{switch\_ends}(n, n')$ denoting that n' is a possible switch destination from n, and $\texttt{at}(s) = n$ to insure the victim is at the relevant node. The effect would be $\texttt{switch}(n, s) = n'$.

(*c*) The victim now wishes to solve their planning problem by converting it to a *constraint satisfaction problem (CSP)*. Explain in detail, giving examples based on the diagram above and your earlier answers, the steps necessary to perform the conversion. [7 marks]

Answer: This essentially requires a description of the entire conversion. At each time step t there will be a CSP variable $\texttt{action}^t$ that can be assigned values such as $\texttt{action}^5 = \texttt{move}(n_{3,4})$. State-variables have CSP variables that model their values at different times, so for example we might have a CSP variable $\texttt{at}^t$ for each time step t, that can be assigned values such as $\texttt{at}^5 = n_{3,4}$. We then need constrains representing preconditions, such as a constraint that when $\texttt{action}^5 = \texttt{move}(n_{1,2})$ we must also have an assignment such as $\texttt{at}^5 = n_{1,1}$. We need constraints modelling the effects of actions, such as requiring $\texttt{action}^5 = \texttt{move}(n_{1,2})$ to appear with $\texttt{at}^6 = n_{1,2}$. Finally we need frame constraints to maintain the values of variables not affected by an action. For example, a three-way constraint requiring $\texttt{action}^5 = \texttt{move\_switch}(\texttt{Start}, n_{1,1})$ to appear with $\texttt{switch}^5(n_{4,3}) = n_{4,4}$ and $\texttt{switch}^6(n_{4,3}) = n_{4,4}$.